

IEM KOLKATA
Advanced Aptitude & Logic
Quantitative Aptitude

Q1) $4^{61} + 4^{62} + 4^{63} + 4^{64} + 4^{65}$ is divisible by?

(a) 3 (b) 5 (c) 11 (d) 17

Q2) Simplify

$$\frac{0.72 \times 0.72 \times 0.72 - 0.39 \times 0.39 \times 0.39}{0.72 \times 0.72 + 0.72 \times 0.39 + 0.39 \times 0.39}$$

Q3) $32^5 + 2^{27}$ is divisible by

(a) 3 (b) 7 (c) 10 (d) 11

Q4) Find the smallest number which when added to 4519 makes it exactly divisible by 30.

(a) 6 (b) 11 (c) 10 (d) 13

Q5) Find the smallest number which on adding 19 to it is exactly divisible by 28, 36 and 45.

(a) 1260 (b) 1241 (c) 1620 (d) 1214

Q6) How many numbers are there between 1 & 900 which are not divisible by 2, 3 or 5?

(a) 240 (b) 245 (c) 250 (d) None

Q7) What is the remainder when $(17^{25} + 19^{25})$ is divided by 18?

(a) 0 (b) 1 (c) 3 (d) 9

Q8) Instead of multiplying a number by $8/3$, a student divided it by $8/3$. What will be the error percentage?

(a) 54% (b) 85.9% (c) 66.66% (d) 76%

Q9) If price of rice is 30% less than that of wheat, then price of wheat is how much per cent more than that of rice?

Q10) The population of a district is 362000, out of which 181000 are males, 47% of the population is literate. If 35% males are literates, then what percentage of females are literate?

(a) 58% (b) 61% (c) 59% (d) 62%

Q11) Find A:B:C:D when A:B=2:3, B:C=7:9, C:D=5:7

Q12) If A:B=11:7, B:C=5:19, then A:B:C=?

(a) 55:35:133 (b) 35:55:133 (c) 35:133:55 (d) 55:133:35

Q13) Ramesh and Mahesh work in a shop and Ramesh's salary is $\frac{5}{6}$ th of the salary of Mahesh. They spend same money of Rs 2000 and after that save all the money. Find the salary of Ramesh and Mahesh if the ratio of their savings is 4 : 5.

(a) 10000, 12000 (b) 15500, 12500 (c) 8000, 10000 (d) 11000, 8000

Q14) Income ratio of Ramesh and Suresh is 5:6. Their spending ratio is 7:9. Ramesh saves 4000 and Suresh saves 3000. The spending amount of Ramesh and Suresh are?

(a) 21000, 27000 (b) 12000 , 18000 (c) 30000 , 36000 (d) 15000 , 21000

Q15) Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

(a) 2 : 3 : 4 (b) 6 : 7 : 8 (c) 6 : 8 : 9 (d) 5 : 7 : 9